

FIRE TEST extended after the button release (flyPad option)

A Realism option on both aircraft: the FIRE TEST goes on for 7 s after the button is released, all its warnings and lights together

Branch	pr/30-fire-test-extension
Aircraft	A32NX (FWC) and A380X (fire protection system, FWS)
Commits	A32NX 2180b3b98 bc012b81d 339adb87b 62edf66ec ; A380X 6fd19fc3c ; docs 088ef54e0
Base	upstream master
Status	CI checks pass; the A32NX part tested in the sim (2026-09-17), the A380X part tested in the sim by the user: OK (2026-09-29).

Purpose

On the aircraft the FIRE TEST stops the instant the button is released; in the simulator the crew must hold it with the mouse while checking the ECAM, the CRC and every light. An unrealistic flyPad option (on by default) keeps the test 7 s after the release.

Specification

Aircraft	Implementation
A32NX	FWC monostables written every tick; the extended APU / ENG 1 / ENG 2 test drives the CRC, MASTER WARN, ECAM, FIRE pb and AGENT / SQUIB DISCH lights (L:A32NX_FWC_FIRE_TEST_*_ACTIVE)
A380X	Rust fire protection system: DelayedFalseLogicGate of 7 s on the FIRE TEST pb before its 500 ms delay, so the fire detection zones, FIRE and squib lights and FWS warnings stay together; L:A32NX_FIRE_TEST_EXTEND (from the systems host), L:A32NX_FIRE_TEST_ACTIVE (to the FWS)

Tests

Rust: the test stops on release without the option, lasts 7 s with it (35 fire tests pass).

PR description (from the PR draft)

From `pr-drafts/30-fire-test-extension.md`, section "Summary of Changes" onwards.

Summary of Changes

A flyPad Settings > Realism option, "Extend Fire Test Warnings After Button Release" (marked unrealistic, on by default): the FIRE TEST goes on for 7 s after the test button is released, so the crew can check the warnings and the lights without holding it. Off, the test stops on release as on the aircraft.

- **A32NX:** the FWC keeps the APU / ENG 1 / ENG 2 FIRE TEST active 7 s after release (monostables written every tick, so no falling edge is missed while the option is off), which keeps the CRC, MASTER WARN, the ECAM and the FIRE pb lights; the AGENT / SQUIB DISCH lights read the same extended signal (`L:A32NX_FWC_FIRE_TEST_{ENG1|ENG2|APU}_ACTIVE`). Also fixes a short-circuit that dropped the APU FIRE TEST.
- **A380X:** the fire protection system (Rust) extends the FIRE TEST pb signal by 7 s before its 500 ms test delay, so the fire detection zones, the FIRE lights, the squib lights and the FWS warnings all stay on together. The systems host passes the option (`L:A32NX_FIRE_TEST_EXTEND`); the FWS reads the extended test (`L:A32NX_FIRE_TEST_ACTIVE`) where it read the pushbutton.
- 2 Rust tests (`fire_and_smoke_protection.rs`: the test stops on release without the option, lasts 7 s with it).

Cockpit API Changes

- A32NX: `L:A32NX_FWC_FIRE_TEST_{ENG1|ENG2|APU}_ACTIVE` (documented in `a320-simvars.md`).
- A380X: `L:A32NX_FIRE_TEST_EXTEND`, `L:A32NX_FIRE_TEST_ACTIVE` (documented in `a380-simvars.md`).

Screenshots (if necessary)

TO ADD: the flyPad Realism option.

References

Not an aircraft behaviour: a realism option for the simulator, like the other unrealistic flyPad options.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. flyPad Settings > Realism: "Extend Fire Test Warnings After Button Release" on.
2. A32NX: press and release ENG 1 FIRE TEST: the CRC, MASTER WARN, ECAM, FIRE pb and AGENT lights stay 7 s, together.
3. A380X: press and release FIRE TEST on the overhead: FIRE and squib lights and the FWS warnings stay 7 s, together.
4. Option off: everything stops on release, on both aircraft.
5. `cargo test -p a380_systems fire_and_smoke`.